Rankstack

Contest participation, evaluated submissions, and ranked results.

rankstack.upayan.dev

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5
MongoDB collections

112
Fresh local sample documents

5
Aggregation pipelines

ARCHITECTURE

Records and live state

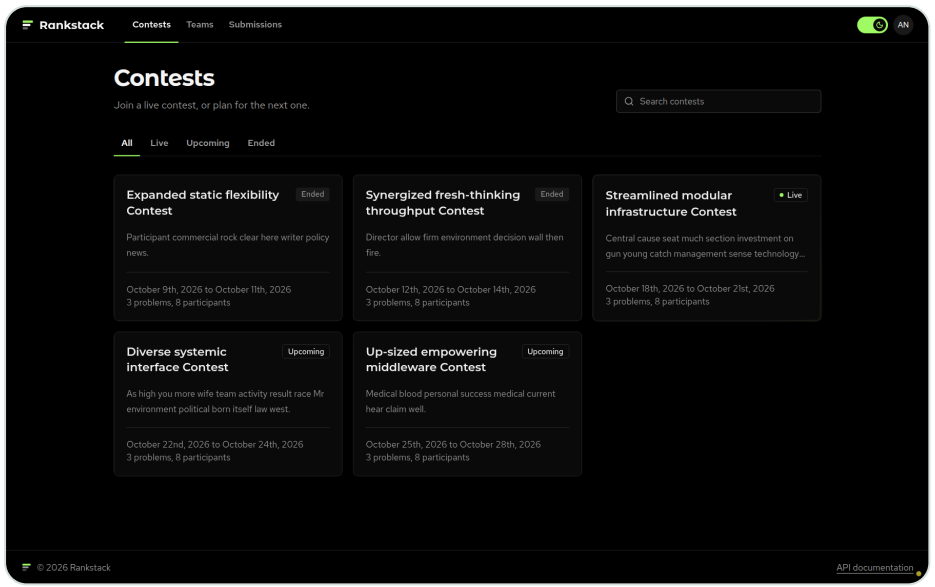
- Next.js provides contest, team, and submission screens.
- FastAPI validates requests and writes related documents in MongoDB transactions.
- Redis stores session keys, rate-limit counters, and leaderboard sorted sets.

MongoDB
Users · teams · contests · problems · submissions

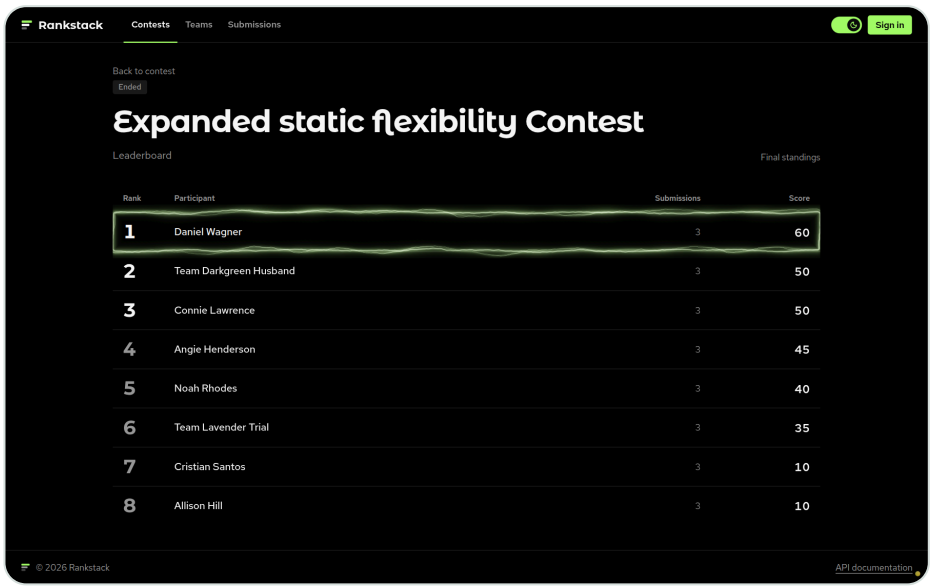
Verified demo
CRUD, two indexed queries, transactions, five aggregations, and Redis operations

WORKING APPLICATION

Contest catalog and standings



Seeded local contest catalog



Seeded local final standings

ADVANCED QUERIES

Five aggregation questions

- Who leads a contest by score?
- How do scores differ by problem difficulty?
- Which verdicts appear in each contest?
- How many distinct problems did each user solve?
- How many submissions and members does each team have?
- Indexed lookups use `IXSCAN` for problems and participant submissions.

DEMONSTRATION

Show the system, then the data

1. Open the website and sign in to the local seeded demo.
2. Browse contests, problems, teams, and the leaderboard.
3. Update a submission verdict as an administrator.
4. Run `make demo` against the seed database.
5. Explain transactions, indexes, and Redis expiration.

The public contest catalog was empty at screenshot time. The local seed supplies 112 documents for the review.